

A Double Category Theoretic Perspective on Partiality

Work in progress

Adjoint School 2026 Group 2

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Introduction

Theorem Cockett and Garner 2014

Restriction categories are categories enriched in $\mathbb{P}\text{StabLat}^{\text{op}}$.

Outline

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Restriction categories

Many frameworks where the notion of a "partially defined" arrow needs to be studied, e.g. sets and partial functions. Ordinary categories cannot capture such notion; we need extra structure to capture "partiality".

Definition [Cockett and Lack 2002]

A **restriction category** is a category $\mathcal{C}$ equipped with a **restriction structure**, that is an operator $\mathcal{C}(c, d) \rightarrow \mathcal{C}(c, c) : f \mapsto \bar{f}$, for all $c, d \in \mathcal{C}_0$ so that

- (R1) $f\bar{f} = f$ for any f ,
- (R2) $\bar{f}\bar{g} = \bar{g}\bar{f}$ for any f, g with $\text{dom}(f) = \text{dom}(g)$,
- (R3) $\overline{f\bar{g}} = \bar{f}\bar{g}$ for any f, g with $\text{dom}(f) = \text{dom}(g)$, and
- (R4) $\bar{h}f = f\bar{h}\bar{f}$ for any f, h with $\text{cod}(f) = \text{dom}(h)$.

Intuitively, $\bar{f}$ is determining where exactly on $\text{dom}(f)$ is f defined, measuring its "degree of partiality". In particular, f will be called **total** when $\bar{f} = 1_{\text{dom}(f)}$.

Examples

1. Any category $\mathcal{C}$ can be trivially made into a restriction category if we define $\bar{f} = 1_c$ for any $f : c \rightarrow d$. All arrows are total.
2. The category Par of sets and partial functions where

$$\bar{f} = \begin{cases} x, & \text{if } f(x) \text{ is defined} \\ \text{undefined,} & \text{otherwise} \end{cases}$$

for any $f : c \rightarrow d$, is a restriction category. The total arrows are the ordinary total functions.

3. The opposite category $\text{StabLat}^{\text{op}}$ meet-semilattices (posets with binary meets and a top element) and stable maps (functions preserving the joins but not necessarily the top element) is a restriction category if we define for any $f : c \leftarrow d$, $\bar{f}$ to be

$$\bar{f}(x) = x \wedge f(\top).$$

Total arrows are those stable maps that also respect the top element.

The fundamental (restriction) functor

For a restriction category $\mathcal{C}$, we can define a partial order on its hom-sets by taking

$$f \leq g \iff f = g\bar{f},$$

intuitively meaning that f and g agree whenever f is defined. This partial order is compatible with the composition making $\mathcal{C}$ into a 2-category.

Given two restriction categories $\mathcal{C}, \mathcal{D}$, a **restriction functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a functor that respects the restriction structure, that is $F\bar{f} = \overline{Ff}$ for any arrow f . In other words, if $\mathcal{O}(c) := \{\bar{f} : c \rightarrow c \mid f \in \mathcal{C}\}$ then F induces a functor $\mathcal{O}(c) \rightarrow \mathcal{O}(Fc)$.

Moreover, $\mathcal{O}(c)$ can be made into a meet-semilattice by taking $\bar{f} \wedge \bar{g} = \bar{f} \circ \bar{g}$ and $\top = 1_c$ and for every arrow $c \rightarrow d$ we get a stable map $\mathcal{O}(d) \rightarrow \mathcal{O}(c)$. In particular, for every restriction category $\mathcal{C}$ we get a restriction functor

$$\mathcal{O} : \mathcal{C} \rightarrow \text{StabLat}^{\text{op}}$$

called the **fundamental functor**.

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Review: double categories

A (weak) double category is a (pseudo-)category internal to $\mathbf{Cat}$:

$$\mathbb{D}_0 \begin{array}{c} \xleftarrow{\text{dom}} \\ \xrightarrow{\text{id}} \\ \xleftarrow{\text{cod}} \end{array} \mathbb{D}_1 \xleftarrow{\odot} \mathbb{D}_1 \times_{\mathbb{D}_0} \mathbb{D}_1$$

Concretely it consists of:

- objects $A, B, \dots$ (objects of $\mathbb{D}_0$);
- tight arrows $f : A \rightarrow B$ (morphisms of $\mathbb{D}_0$);
- loose arrows $f : A \rightharpoonup B$ (objects of $\mathbb{D}_1$);

• cells $\begin{array}{ccc} A & \xrightarrow{+} & B \\ \downarrow & \alpha & \downarrow \\ C & \xrightarrow{+} & D \end{array}$ (morphisms of $\mathbb{D}_1$);

+ identity arrows, composition operations, (associator, unitors), axioms...

Example

Let $\mathcal{V}$ be a distributive monoidal category. The **double category of matrices** $\mathbb{Mat}(\mathcal{V})$ has sets and functions as objects and tight arrows, matrices $M : A \times B \rightarrow \mathcal{V}$ as loose arrows $M : A \rightharpoonup B$, and cells given by

$$\begin{array}{ccc}
 A & \xrightarrow{M} & B \\
 f \downarrow & \alpha & \downarrow g \\
 C & \xrightarrow{N} & D
 \end{array}
 \quad := \quad
 \begin{array}{ccc}
 A \times B & \xrightarrow{M} & \mathcal{V} \\
 f \times g \downarrow & \Downarrow \alpha & \\
 C \times D & \xrightarrow{N} & \mathcal{V}
 \end{array}$$

Composition of loose arrows is given by matrix multiplication:

$$(M \odot N)(a, c) := \bigsqcup_{b \in B} M(b, c) \otimes N(a, b)$$

If $\mathcal{V}$ is the poset $\{0 \leq 1\}$ (with tensor min and unit 1), then $\mathbb{Mat}(\mathcal{V}) = \mathbb{Rel}$ the double category of sets and relations.

Review: Enriched categories

Let $(\mathcal{V}, \otimes, I)$ be a monoidal category. A $\mathcal{V}$ -category $\mathcal{C}$ is given by:

- a set $\text{ob}(\mathcal{C})$ of objects;
- for every pair of objects A, B , a **hom-object** $\mathcal{C}(A, B) \in \mathcal{V}$;
- for every objects A, B, C , morphisms

$$j_A : I \rightarrow \mathcal{C}(A, A) \quad \text{and} \quad \circ_{A,B,C} : \mathcal{C}(B, C) \otimes \mathcal{C}(A, B) \rightarrow \mathcal{C}(A, C)$$

subject to some axioms (composition is associative and unital. . .).

Enriching in a double category

Let $\mathbb{D}$ be a double category. A $\mathbb{D}$ -category $\mathcal{C}$ is given by:

- a set $\text{ob}(\mathcal{C})$ of objects;
- a **function** $|-| : \text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathbb{D})$;
- for every pair of objects A, B , a **loose arrow** $\mathcal{C}(A, B) : |A| \rightarrowtail |B|$ of $\mathbb{D}$;
- for every objects A, B, C , cells

$$\begin{array}{ccc}
 |A| & \xrightarrow{\text{id}_{|A|}} & |A| \\
 \parallel & & \parallel \\
 |A| & \xrightarrow{\mathcal{C}(A,A)} & |A|
 \end{array}
 \quad j_A \quad
 \begin{array}{ccc}
 |A| & \xrightarrow{\mathcal{C}(A,B)} & |B| \\
 \parallel & & \parallel \\
 |A| & \xrightarrow{\mathcal{C}(A,C)} & |C|
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 |A| & \xrightarrow{\mathcal{C}(A,B)} & |B| & \xrightarrow{\mathcal{C}(B,C)} & |C| \\
 \parallel & & \parallel & & \parallel \\
 |A| & \xrightarrow{\mathcal{C}(A,C)} & |C|
 \end{array}
 \quad \circ_{A,B,C}$$

subject to some axioms (composition is associative and unital. . .).

Examples

- Let $\mathcal{V}$ be a monoidal category. We can see $\mathcal{V}$ as a one object and one tight arrow double category $\mathbb{V}$. A $\mathbb{V}$ -category is exactly a $\mathcal{V}$ -category.
Conclusion: Categories enriched in double categories are direct generalizations of monoidal enriched categories.
- A $\mathbf{Mat}(\mathcal{V})$ -category $\mathcal{C}$ is a collection $(X_A)_{A \in \text{ob}(\mathcal{C})}$ of sets, and for each pair X_A, X_B a matrix

$$\mathcal{C}(A, B) : X_A \times X_B \rightarrow \mathcal{V} \quad + \text{ conditions}$$

Intuitively: $\mathcal{C}(A, B)$ can have many “inputs” and “outputs”.

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Recovering restriction structure I

Reminder: Every restriction category $\mathcal{C}$ has its fundamental functor $\mathcal{O}_{\mathcal{C}}: \mathcal{C} \rightarrow \text{StabLat}^{\text{op}}$

Idea: Can we define restriction categories in terms of $\mathcal{O} = \mathcal{O}_{\mathcal{C}}$? I.e. do restriction categories correspond to functors into $\text{StabLat}^{\text{op}}$?

Not in general, but we know more about $\mathcal{O}$:

- $\mathcal{O}$ can be viewed as a 2-functor $\mathcal{C} \rightarrow \text{StabLat}^{\text{op}}$ (it preserves the ordering of morphisms)
- $\mathcal{O}$ is a local discrete fibration ($\mathcal{O}: \mathcal{C}(c, c') \rightarrow \text{StabLat}^{\text{op}}(\mathcal{O}(c), \mathcal{O}(c'))$ is a discrete fibration for every $c, c' \in \mathcal{C}$)

Recovering restriction structure II

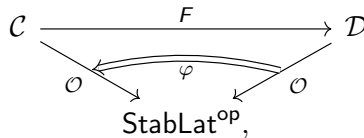
Theorem [Cockett and Garner 2014]

Every local discrete fibration $F: \mathcal{D} \rightarrow \text{StabLat}^{\text{op}}$ determines a unique restriction structure on $\mathcal{D}$ such that $\mathcal{O}_{\mathcal{D}} = F$.

What about restriction functors?

Theorem [Cockett and Garner 2014]

Restriction functors $F: \mathcal{C} \rightarrow \mathcal{D}$ correspond to pairs of a 2-functor $F: \mathcal{C} \rightarrow \mathcal{D}$ together with a 2-natural transformation φ



whose components are discrete fibrations.

Local discrete fibrations as enriched categories

Theorem [Cockett and Garner 2014]

For every 2-category $\mathcal{K}$, there is a double category $\mathbb{P}\mathcal{K}$ such that the 2-category of $\mathbb{P}\mathcal{K}$ -categories is 2-equivalent to $2\text{-Cat} //_{\text{Idf}} \mathcal{K}$.

Corollary [Cockett and Garner 2014]

The 2-category of restriction categories is 2-equivalent to the 2-category of $\mathbb{P}(\text{StabLat}^{\text{op}})$ -categories.

The construction $\mathbb{P}$ is coming out of nowhere, can we conceptualise it?

Proposition [our group]

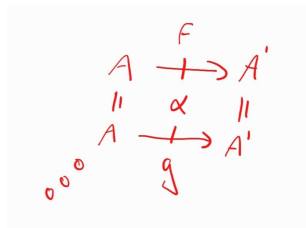
$$\mathbb{P}\mathcal{K} = \widehat{\mathbb{D}fib\mathcal{K}}$$

Local cocompletion of double categories

Definition [our group]

For every double category $\mathbb{D}$, the double category $\hat{\mathbb{D}}$ has

- objects... objects of $\mathbb{D}$
- tight arrows... tight arrows of $\mathbb{D}$
- loose arrows $A \rightarrowtail A'$... functors $\mathbb{D}(A, A')^{\text{op}} \rightarrow \text{Set}$
- cells... too technical



Conjecture

$\hat{\mathbb{D}}$ is the *local cocompletion* of $\mathbb{D}$.

Lax slice of a double category

Theorem [Fujii and Lack 2024]

Given a bicategory $\mathcal{B}$ and a $\mathcal{B}$ -category $\mathcal{C}$, there is a bicategory $\mathcal{B}/\mathcal{C}$ such that $(\mathcal{B}/\mathcal{C})\text{-Cat} \simeq \mathcal{B}\text{-Cat}/\mathcal{C}$.

Theorem [our group]

Given a double category $\mathbb{D}$ and a $\mathbb{D}$ -category $\mathcal{C}$, there is a double category $\mathbb{D}//\mathcal{C}$ such that $(\mathbb{D}//\mathcal{C})\text{-Cat} \simeq \mathbb{D}\text{-Cat}//\mathcal{C}$.

Proposition [our group]

The double category $\mathbb{P}\text{StabLat}^{\text{op}}$ is a sub-double category of $\Sigma\text{Cat}//\text{StabLat}^{\text{op}}$.

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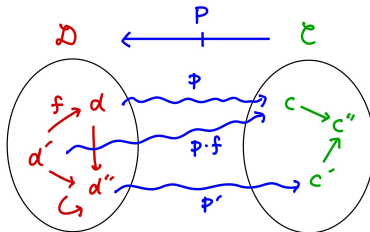
Correspondence of distributors

Distributors

Definition

A **distributor** $P: \mathcal{C} \rightarrow \mathcal{D}$ (a.k.a. profunctor, bimodule) for categories $\mathcal{C}, \mathcal{D}$ is a functor $\mathcal{D}^{\text{op}} \times \mathcal{C} \rightarrow \text{Set}$.

Intuitively, the set $P(d, c)$ is a collection of “morphisms” $p: d \rightsquigarrow c$.



sets :: functions :: relations $\approx$ *categories :: functors :: distributors*

Double categories of distributors

Categories, functors, distributors, and transformations form a double category $\mathbb{D}\text{ist}$.

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{P} & \mathcal{B} \\
 F \downarrow & \varphi & \downarrow G \\
 \mathcal{C} & \xrightarrow{Q} & \mathcal{D}
 \end{array}
 \quad \parallel \quad
 \begin{array}{ccc}
 \mathcal{B}^{\text{op}} \times \mathcal{A} & \xrightarrow{P} & \text{Set} \\
 G^{\text{op}} \times F \downarrow & \Downarrow \varphi & \\
 \mathcal{D}^{\text{op}} \times \mathcal{C} & \xrightarrow{Q} &
 \end{array}$$

Similarly, for any double category $\mathbb{D}$, we get a **virtual** double category $\mathbb{D}\text{-}\mathbb{D}\text{ist}$ of categories enriched in $\mathbb{D}$.

These **virtual** double categories can be thought of as stages for category theory, where categorical notions (weighted limits, cocompletion, etc.) can be defined and studied.

Restriction distributors

Definition [DeWolf 2017]

A **restriction distributor** $P: \mathcal{C} \multimap \mathcal{D}$ for restriction categories $\mathcal{C}, \mathcal{D}$ is a distributor $\mathcal{C} \multimap \mathcal{D}$ equipped with an operator $\overline{(-)}: P(d, c) \rightarrow \mathcal{O}(c)$ such that

- (RD1) $p\overline{p} = p$ for any $p: d \rightsquigarrow c$,
- (RD2) $\overline{p\overline{f}} = \overline{p}\overline{f}$ for any $p: d \rightsquigarrow c$ and $f: d \rightarrow d'$,
- (RD3) $\overline{p}f = f\overline{p}f$ for any $f: d' \rightarrow d$ and $p: d \rightsquigarrow c$, and
- (RD4) $\overline{g}p = p\overline{g}p$ for any $p: d \rightsquigarrow c$ and $g: c \rightarrow c'$.

$\overline{(-)}$ is the operator on “morphisms” $p: d \rightsquigarrow c$ that looks like the restriction operator of a restriction category.

Equivalence of the notion of distributors

Theorem [our group]

Restriction distributors $\mathcal{C} \rightarrow \mathcal{D}$ correspond to distributors enriched in $\mathbb{P}\text{StabLat}^{\text{op}}$.

Conjecture

By defining the double category $\mathbb{R}\text{Dist}$ of restriction categories, restriction functors, and restriction distributors, with a suitable notion of transformations, we would get

$$\mathbb{R}\text{Dist} \simeq \mathbb{P}\text{StabLat}^{\text{op}}\text{-Dist}$$

This would improve the equivalence of the 2-categories of restriction categories and $\mathbb{P}\text{StabLat}^{\text{op}}$ -enriched categories proved in [Cockett and Garner 2014].

Conclusion

Summary

- We found two new presentations of the double category, in which enriched categories are restriction categories.
- We extended this to an equivalence of distributors.

Future work

- Prove the conjectures.
- Study the two constructions of **virtual** double categories $\widehat{(-)}$ and $(-)//(-)$ in more detail.

References I

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Thank you!